

RAG Chatbot Sample Document

This document exists to test the retrieval augmented generation pipeline. The capital of France is Paris. The Eiffel Tower is located in Paris. MongoDB Atlas Vector Search stores high dimensional embedding vectors and supports approximate nearest neighbor search using the cosine similarity metric. Google Gemini provides the text-embedding-004 model which outputs 768 numbers per chunk of text. The chat model used in this project is gemini-2.0-flash.

Retrieval augmented generation works in three stages. First, documents are split into overlapping chunks. Second, each chunk is embedded into a vector and stored in a database. Third, at query time the user question is embedded and the most similar chunks are retrieved and passed to the language model as context.

The recommended chunk size for this project is one thousand characters with an overlap of two hundred characters. Overlap preserves context across boundaries. This sample text is intentionally long enough to produce more than one chunk so that splitting behavior can be observed clearly during the seeding step.

Frequently asked question: What database does this project use for vectors?
Answer: It uses MongoDB Atlas Vector Search. Frequently asked question: which embedding model is used? Answer: Google Gemini text-embedding-004 with 768 dims.